Lab 1. Local APIs & The Power of Abstraction (npm)

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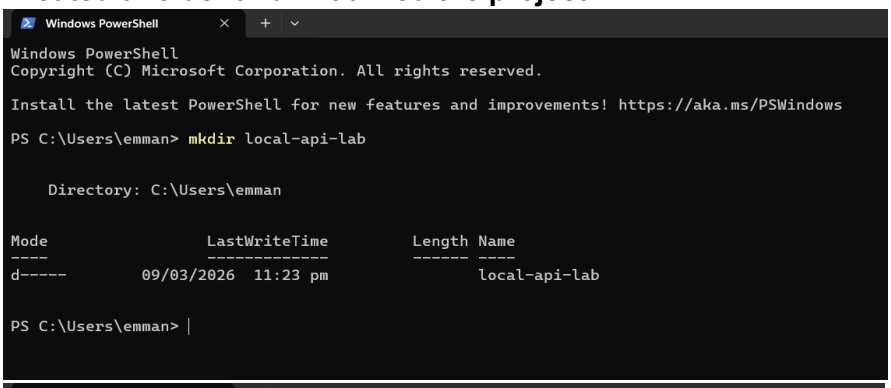
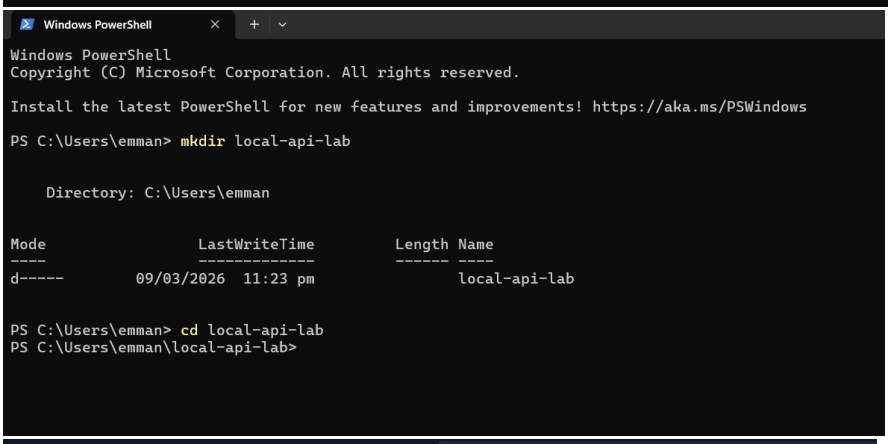
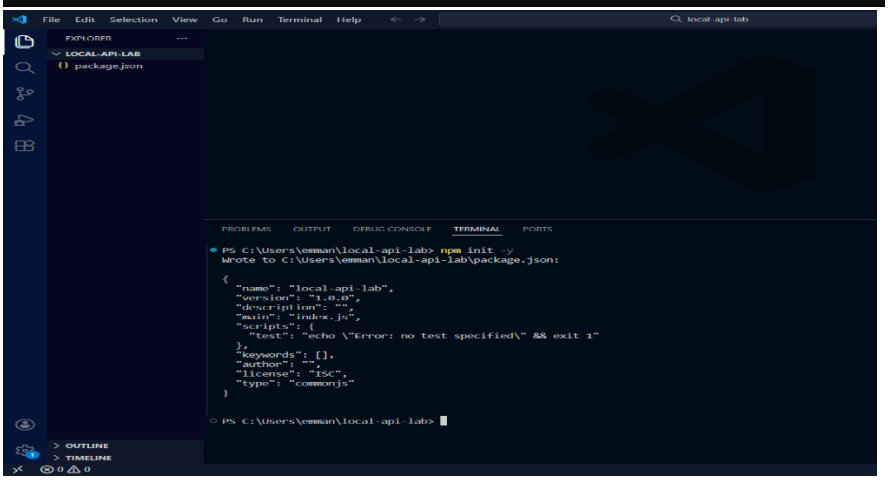
Instructions:

1. **Initialize a Project:** Open your terminal, create a new folder named local-api-lab, navigate into it, and run the command npm init -y. This creates a package.json file to manage your local modules.

   ✓ **Created a folder and initialized the project.**

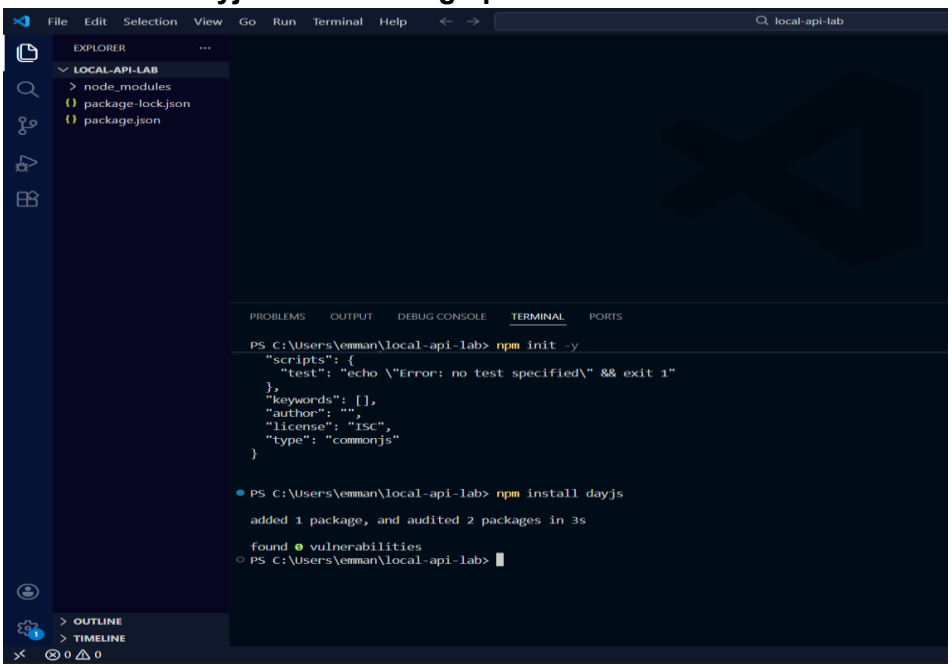






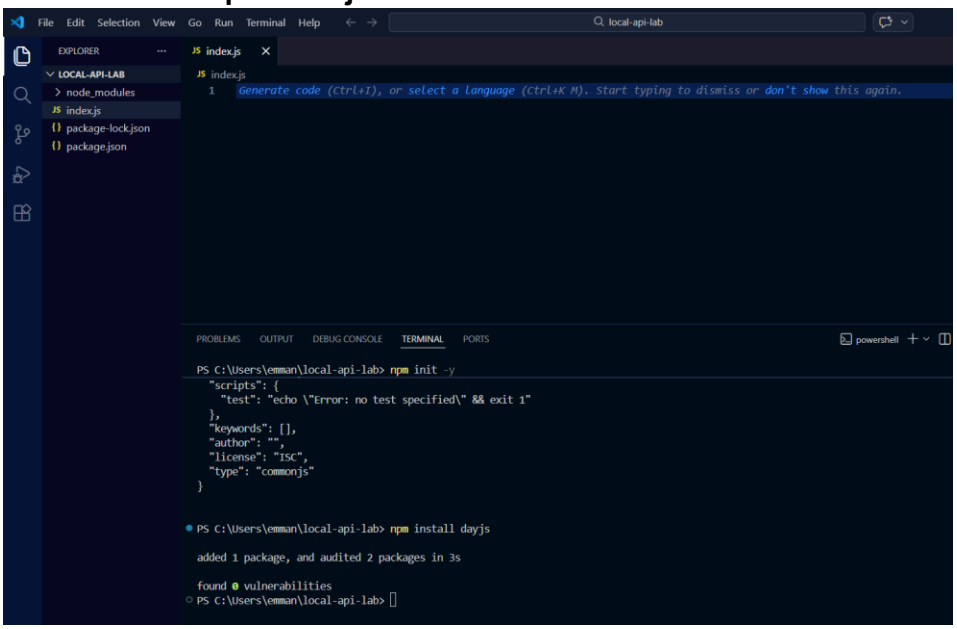
2. **Install a Module:** We will use a date-formatting module so you don't have to write complex calendar math from scratch. Run: npm install dayjs

✓ **Installed the dayjs module using npm**



3. **Use the Local API:** Create a script to interact with the dayjs API

✓ **Created the script index.js**



a. "Summon" the module you installed via npm
   i. use the 'require' function

```
JS index.js > ...
1   const dayjs = require ('dayjs');
```

b. Capture the current moment

```
JS index.js > ...
1   const dayjs = require ('dayjs');
2
3   const now = dayjs();
```

c. Transform 'now' into a readable string
   i. You need a method that takes a string like 'MMMM D, YYYY'

```
JS index.js > ...
1   const dayjs = require ('dayjs');
2
3   const now = dayjs();
4
5   const currentDate = now.format('MMMM D, YYYY');
```

d. Do the calendar math
   i. Use a method to 'add' a value of 7 and the unit 'day'

```
JS index.js > ...
1   const dayjs = require ('dayjs');
2
3   const now = dayjs();
4
5   const currentDate = now.format('MMMM D, YYYY');
6
7   const nextWeek =now.add(7, 'day').format('MMMM D, YYYY');
```

e. Print current date and date for next week

```
Go  Run  Terminal  Help        local-api-lab
JS index.js
JS index.js > ...
1   const dayjs = require ('dayjs');
2
3   const now = dayjs();
4
5   const currentDate = now.format('MMMM D, YYYY');
6
7   const nextWeek =now.add(7, 'day').format('MMMM D, YYYY');
8
9   console.log("Current Date:", currentDate);
10  console.log("Date Next Week:", nextWeek);

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS        Filter (e.g. text, !exclude, \esca...
C:\Program Files\nodejs\node.exe .\index.js
Current Date: March 9, 2026
Date Next Week: March 16, 2026
```

4. **Question:** How does using dayjs demonstrate the concept of "abstraction" compared to writing a function that calculates leap years and days of the month manually?
   - Using dayjs demonstrates abstraction because it hides the complex logic of date calculations like leap years and different days in each month. Instead of writing the code manually, we simply use built-in methods like format() and add().